

What a Popular PCOS Dataset Misses in the PMOS Era

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Plain-English Summary

PCOS has been renamed polyendocrine metabolic ovarian syndrome (PMOS). The new name matters because it points away from a narrow ovarian-cyst framing and toward a lifelong endocrine and metabolic condition. The Kaggle dataset is useful for classroom statistics and exploratory modeling, but it was built around the older PCOS measurement frame.

What the Dataset Covers

Table 1: PMOS domain coverage in the Kaggle dataset.

PMOS domain	Coverage	Major gaps
Reproductive / ovarian	Strong	diagnostic visit notes and formal phenotype labels
Endocrine	Partial	testosterone, SHBG, DHEAS, free androgen index
Metabolic / cardiovascular	Weak proxy	fasting insulin, fasting glucose, HOMA-IR, OGTT, lipids
Dermatological / symptoms	Partial	standardized hirsutism/acne scales
Psychological / quality of life	Absent	depression, anxiety, sleep, stigma, quality-of-life scales

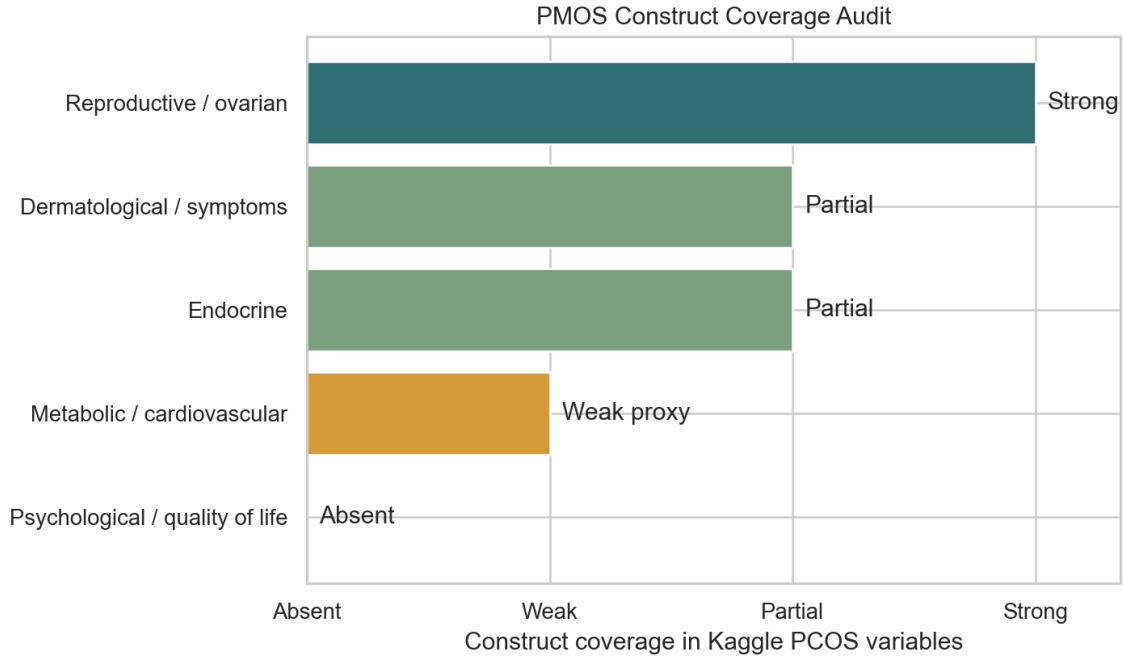


Figure 1: The dataset is strongest on reproductive/ovarian variables and weakest on metabolic and psychological PMOS domains.

What the BMI Analysis Adds

In the PCOS-positive subset, mean BMI was 25.47. The clinical-only BMI model had adjusted $R^2 = 0.257$, while the laboratory-only model had adjusted $R^2 = -0.008$. The clearest predictor was self-reported weight gain.

This is not evidence that hormones or metabolism are unimportant. It is evidence that this public dataset is missing direct metabolic biosignals such as fasting insulin, fasting glucose, HOMA-IR, OGTT, lipids, and testosterone.

Classification Context

We also ran two simple logistic classifiers for PCOS/PMOS status. These are not diagnostic tools; they show what kind of signal the dataset contains.

Table 2: Classifier context results.

Model	Test n	AUC	Brier score	Sensitivity	Specificity
Ovary-centric	162	0.919	0.106	0.792	0.899
Metabolic + symptom	162	0.860	0.135	0.736	0.881

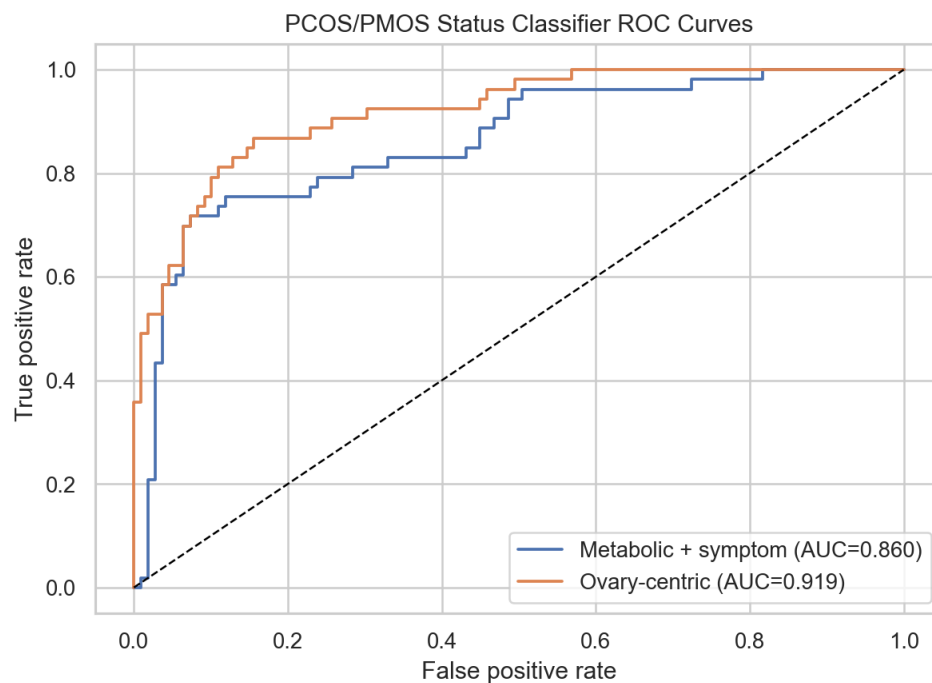


Figure 2: ROC curves for two PCOS/PMOS-status classifiers.

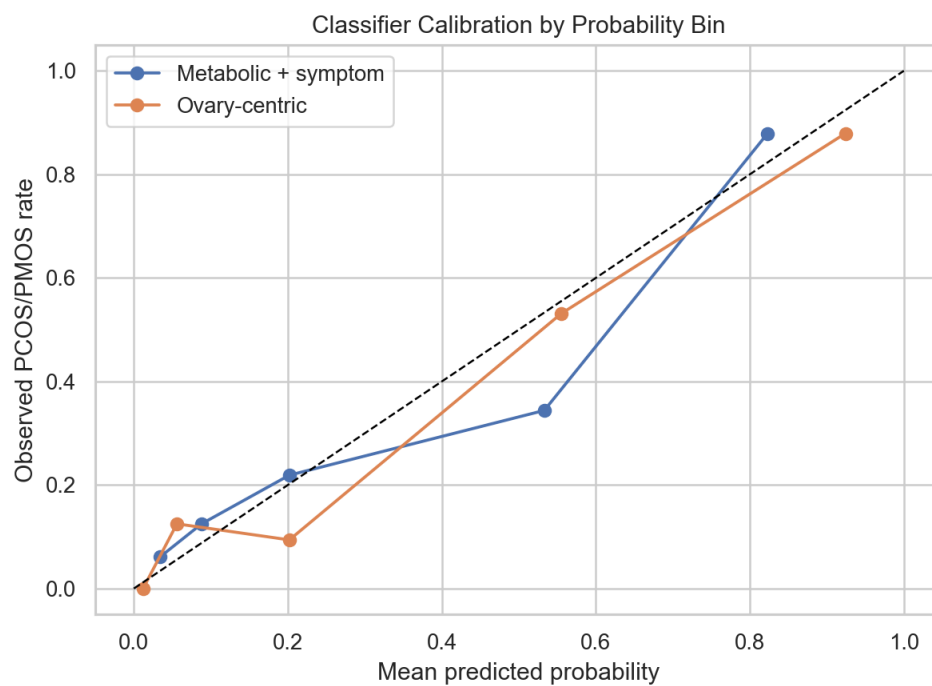


Figure 3: Calibration curves by predicted-probability bin.

Public Takeaway

A dataset can only teach what it measures. This Kaggle dataset can support exploratory PCOS-status modeling, but it cannot fully represent PMOS as a polyendocrine metabolic condition. Future PMOS

datasets should include fasting insulin, fasting glucose, HOMA-IR, lipids, blood pressure, androgen assays, mental-health scales, and longitudinal follow-up.

Source Links

- Endocrine Society PMOS release: <https://www.endocrine.org/news-and-advocacy/news-room/2026/pcos-name-change>
- Monash PMOS release: <https://www.monash.edu/medicine/news/latest/2026-articles/polyendocrine-metabolic-ovarian-syndrome-new-name-to-improve-diagnosis-and-care-of-condition-affecting-170-million-women-worldwide>
- Kaggle dataset: <https://www.kaggle.com/datasets/prasoonkottarathil/polycystic-ovary-syndrome-pcos>